



(<https://docs.aws.amazon.com>)



Get started  
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Documentation

(<https://docs.aws.amazon.com/index.html>)



AWS Identity and Access Management

(<https://docs.aws.amazon.com/iam/index.html>)



User G...

# What is IAM?

↓ PDF (</pdfs/IAM/latest/UserGuide/iam-ug.pdf#introduction>)

↓ RSS (<aws-iam-release-notes.rss>)

↓ Markdown (<introduction.md>)

⚡ Agent Setup



Focus mode

AWS Identity and Access Management (IAM) is a web service that helps you securely control access to AWS resources. With IAM, you can manage permissions that control which AWS resources users can access. You use IAM to control who is authenticated (signed in) and authorized (has permissions) to use resources. IAM provides the infrastructure necessary to control authentication and authorization for your AWS accounts.

For instructions on setting up your AWS account and creating identities, see [Getting started with IAM](#) ([./getting-started.html](/getting-started.html)) .

## Identities

When you create an AWS account, you begin with one sign-in identity called the AWS account *root user* that has complete access to all AWS services and resources. We strongly recommend that you don't use the root user for everyday tasks. For tasks that require root user credentials, see [Tasks that require root user credentials](#) ([https://docs.aws.amazon.com/IAM/latest/UserGuide/id\\_root-user.html#root-user-tasks](https://docs.aws.amazon.com/IAM/latest/UserGuide/id_root-user.html#root-user-tasks)) in the *IAM User Guide*.

Use IAM to set up other identities in addition to your root user, such as administrators, analysts, and developers, and grant them access to the resources they need to succeed in

their tasks.

## Access management

After a user is set up in IAM, they use their sign-in credentials to authenticate with AWS. Authentication is provided by matching the sign-in credentials to a principal (an IAM user, AWS STS federated principal, IAM role, or application) trusted by the AWS account. Next, a request is made to grant the principal access to resources. Access is granted in response to an authorization request if the user has been given permission to the resource. For example, when you first sign in to the console and are on the console Home page, you aren't accessing a specific service. When you select a service, the request for authorization is sent to that service and it looks to see if your identity is on the list of authorized users, what policies are being enforced to control the level of access granted, and any other policies that might be in effect. Authorization requests can be made by principals within your AWS account or from another AWS account that you trust.

Once authorized, the principal can take action or perform operations on resources in your AWS account. For example, the principal could launch a new Amazon Elastic Compute Cloud instance, modify IAM group membership, or delete Amazon Simple Storage Service buckets.

### Tip

AWS Skill Builder provides a 10-minute video introduction to IAM:

[Introduction to AWS Identity and Access Management](https://skillbuilder.aws/learn/M1QWQ1MURQ/introduction-to-aws-identity-and-access-management-iam/W4W2NQF2AR) 

(<https://skillbuilder.aws/learn/M1QWQ1MURQ/introduction-to-aws-identity-and-access-management-iam/W4W2NQF2AR>) on the AWS Skill Builder website.

## Service availability

IAM, like many other AWS services, is [eventually consistent](https://wikipedia.org/wiki/Eventual_consistency)  ([https://wikipedia.org/wiki/Eventual\\_consistency](https://wikipedia.org/wiki/Eventual_consistency)). IAM achieves high availability by replicating data across multiple servers within Amazon's data centers around the world. If a request to change some data is successful, the change is committed and safely stored. However, the change must be replicated across IAM, which can take some time. Such changes include creating or updating users, groups, roles, or policies. We recommend that you do not include such IAM changes in the critical, high-availability code paths of your application. Instead, make IAM changes in a separate initialization or setup routine that you run less frequently. Also, be sure to verify that the changes have been propagated before production workflows depend on them. For more information,

see [Changes that I make are not always immediately visible](#) ([./troubleshoot.html#troubleshoot\\_general\\_eventual-consistency](#)) .

## Service cost information

AWS Identity and Access Management (IAM), AWS IAM Identity Center and AWS Security Token Service (AWS STS) are features of your AWS account offered at no additional charge. You are charged only when you access other AWS services using your IAM users or AWS STS temporary security credentials.

IAM Access Analyzer external access analysis is offered at no additional charge. However, you will incur charges for unused access analysis and customer policy checks. For a complete list of charges and prices for IAM Access Analyzer, see [IAM Access Analyzer pricing](#) [↗](https://aws.amazon.com/iam/access-analyzer/pricing) (<https://aws.amazon.com/iam/access-analyzer/pricing>) .

For information about the pricing of other AWS products, see the [Amazon Web Services pricing page](#) [↗](https://aws.amazon.com/pricing/) (<https://aws.amazon.com/pricing/>) .

## Integration with other AWS services

IAM is integrated with many AWS services. For a list of AWS services that work with IAM and the IAM features the services support, see [AWS services that work with IAM](#) ([./reference\\_aws-services-that-work-with-iam.html](#)) .

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## Related resources

[AWS Identity and Access Management API Reference](#)  
(<https://docs.aws.amazon.com/IAM/latest/APIReference/index.html>)

[AWS CLI commands for AWS Identity and Access Management](#)  
(<https://docs.aws.amazon.com/cli/latest/reference/iam/>)

[SDKs & Tools](#) [↗](https://aws.amazon.com/tools/) (<https://aws.amazon.com/tools/>)

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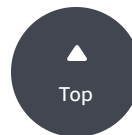
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## Get Started

[AWS Hands-On Tutorials](#) 

(<https://aws.amazon.com/getting-started/hands-on/>)

[AWS Solutions Library](#) 

(<https://aws.amazon.com/solutions/>)

[AWS Decision Guides](#)

(<https://docs.aws.amazon.com/decision-guides/>)

## Developer Tools

[AWS Code Example Library](#)

(<https://docs.aws.amazon.com/code-library/latest/ug/what-is-code-library.html>)

[AWS CLI](#) (<https://docs.aws.amazon.com/cli/>)

[AWS Builder Center](#) 

(<https://builder.aws.com/>)

[AWS Developer Tools Blog](#) 

(<https://aws.amazon.com/blogs/developer/>)

## Service Guides

[Choosing a generative AI service](#)

(<https://docs.aws.amazon.com/decision-guides/latest/decision-guides/genai-guide.html>)

[AWS service guides](#)

(<https://docs.aws.amazon.com/#products>)

[AWS CLI Tutorials on GitHub](#) 

(<https://github.com/aws-samples/sample-developer-tutorials>)

## Helpful Links

[Download the AWS Docs MCP Server](#) 

(<https://github.com/aws-labs/mcp/tree/main/src/aws-documentation-mcp-server>)

[Sign into the AWS Console](#) 

(<https://console.aws.amazon.com/>)

[AWS re:Post](#)  (<https://repost.aws/>)

